MILITARY SPECIFICATION

ANTENNA SYSTEM, BROADBAND IDENTIFICATION FRIEND OR FOE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements of a broadband identification friend or foe (IFF) antenna system. The antenna system operates in two bands: high frequency (1030 MHz ± 10 MHz and 1090 MHz ± 10 MHz), and low frequency (600 MHz to 800 MHz). The high frequency band operates with sum and difference radiated patterns in an ISCS mode at 1030 MHz ± 10 MHz. The low frequency band and the 1090 MHz ± 10 MHz operate in the sum pattern only.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-17/74	Cable, Radio Frequency, Coaxial, RG-213/U
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1631	Procedures For Selection Of Electronic And Electrical Parts During Equipment Design

HANDBOOKS

MILITARY

MIL-HDBK-216	RF Transmission Lines And Fittings

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General. The antenna system shall conform to the requirements of MIL-E-16400 to the extent specified herein.

3.1.1 Applicable requirements of MIL-E-16400. The antenna system shall conform to the following requirements of MIL-E-16400:

a. Salt fog
b. Wind velocity
c. Icing
d. Thermal design
e. Nuclear air blast (7 psi)
f. Shock (high-impact)
g. Vibration
h. Parts, materials, and processes
i. Electrical design and construction
j. Maintenance design
k. Mechanical design and construction (except time totalizing meters are not required)
l. Identification and marking
m. System safety
n. Workmanship

The requirements of a through n shall apply to above-deck equipments, and f through n shall apply to below-deck equipments.

3.1.2 Human engineering. Guidelines in applying human engineering design criteria and principles in the design of the equipment shall be in accordance with Requirement 62 of MIL-STD-454.

3.1.3 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3.

3.2 Parts and materials.

3.2.1 Selection of parts and materials. Parts and materials shall be selected and applied in accordance with MIL-E-16400. Parts shall be derated in usage to provide a minimum safety factor of 100 percent for the critical parameters when practical. Component parts shall be rated at 125°C minimum, or higher if required by Thermal Evaluation of circuit application. Plastic encapsulated micro-circuits shall not be used.

3.2.1.1 Non-standard parts. When it has been determined that design requirements cannot be met by using standard parts or materials specified in MIL-E-16400, the use of non-standard parts shall be in accordance with the Approval Of Non-Standard Parts paragraph of MIL-E-16400.

3.2.1.1.1 Prescreening of non-standard parts. Prescreening of non-standard parts shall be in accordance with MIL-STD-1631.

3.3 Maintainability. The antenna system shall have a mean-corrective-maintenance-time ($\overline{M}_{ct}$) not exceeding 30 minutes when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies and chassis-mounted parts (this includes electronic, electrical, electro-mechanical, and mechanical parts). Corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. The 30 minute $\overline{M}_{ct}$ assumes direct access (colocation) to all subsystems of the antenna system.

3.4 Operating temperature range. The complete broadband antenna system shall be capable of conformance to the Range 2 requirements of MIL-E-16400, except that the high temperature limit shall be 125°C.

3.5 Humidity. The broadband antenna system shall maintain the specified performance when exposed to a relative humidity of 95 percent for both continuous and intermittent periods, including conditions wherein condensation takes place in and on the equipment in the form of both water and frost.

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3.6 Broadband IFF antenna system. The broadband IFF antenna system shall be capable of operating in two bands 1) high frequency, 1030 MHz ± 10 MHz to 1090 MHz ± 10 MHz and 2) low frequency, 600 MHz to 800 MHz. The high frequency band shall operate with sum and difference radiated patterns in an interrogator side lobe suppression (ISLS) mode at 1030 MHz ± 10 MHz. The low frequency band and 1090 MHz ± 10 MHz shall operate in the sum pattern only. The impact impedance (nominal) shall be 50 ohms.

The antenna system shall be composed of a vertically polarized horizontal linear array consisting of the following components, as shown in FIGURE 1:

a. Radiating elements
b. Two feeder circuits
c. Signal processor assembly
d. Backfill radiator assembly
e. Structure assembly
f. Mounting box assembly
g. Filter diplexer

3.7 Printed circuit boards (PCB). Printed circuit boards within each module or assembly shall use plug-in construction to facilitate ease of removal and insertion.

3.8 Thermal design. The antenna system shall conform to the thermal design requirements of MIL-E-16400, including the requirements for thermal survey.

3.9 Antenna electrical design.

3.9.1 Broadband antenna system. The antenna system shall be capable of handling pulses of RF energy of five KW peak at 0.5 percent duty cycle without breakdown.

3.9.2 Gain. The gain of the broadband antenna system, when measured at the sum input to the hybrid, shall be at least 16.0 dB in the frequency band of 600 to 800 MHz. The gain when measured at the hybrid shall be at least 17 dB in the frequencies of 1030 and 1090 MHz.

3.9.3 Beamwidth

3.9.3.1 Horizontal beamwidth. The horizontal beamwidth shall be no greater than 12.5 degrees at 600 MHz and no greater than 9.5 degrees at 800 MHz. The beamwidth shall be no greater than 7.5 degrees at 1030 MHz and at 1090 MHz.

3.9.3.2 Horizontal difference (1030 ± 10 MHz). The horizontal difference pattern shall be sufficient to cover all sidelobes and backlobes by 4.0 dB or greater. The depth of the difference pattern null shall be at least 30 dB below the sum pattern peak at 0°C.

3.9.3.3 Vertical beamwidth. The vertical sum pattern beamwidth shall be 55 to 65 degrees in the frequency band of 600 to 800 MHz. The vertical beamwidth shall be 55 to 65 degrees at 1030 MHz and 1090 MHz.

3.9.3.4 Sidelobes. The horizontal and vertical pattern sidelobes shall be at least -20 dB from the main lobe at all frequencies; that is, 1030 MHz, 1090 MHz, and 600 to 800 MHz.

3.9.3.5 Backlobes. The backlobe level in both the horizontal and vertical planes shall be -25 dB except at 1030 MHz and 1090 MHz. At 1030 MHz and 1090 MHz the backlobe shall be at least -15 dB.

3.9.3.6 VSWR. The antenna system shall have a VSWR of less than 1.4:1 from 600 MHz to 800 MHz and at 1030 ± 10 MHz and 1090 ± 10 MHz.

3.9.4 Connectors and cables. RF Type N connectors shall be used and shall be selected from MIL-HDBK-216. Cables shall be in accordance with MIL-C-17/74. All necessary mating connectors shall be provided as part of the equipment.

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3.9.5 Antenna mechanical design.

3.9.5.1 Weights and dimensions. The broadband antenna system shall not exceed a maximum weight of 65.77 kilograms (145 pounds). The maximum dimensions shall be 3175.0 mm (125 inches) long, 711.2 mm (28 inches) high, and 914.4 mm (36 inches) deep.

3.9.5.2 Antenna structure assembly and mounting box assembly. The antenna structure assembly and the mounting box assembly shall be an integral part of the antenna. The mounting box assembly shall be basically a one-piece construction, symmetrical about the antenna centerline and extending to the center of each moment arm of the antenna. The design of the antenna shall be such that it shall be possible to feed the antenna directly by coaxial cables through approved RF connectors. The electrical characteristics of the antenna and feed system shall not be adversely affected when the antenna is fed in this manner.

3.9.5.3 Structure assembly. The structure assembly shall be a one-piece construction with provisions for adequately supporting the radiating elements, feeder circuits, signal processor assembly, and backfill radiator assembly.

3.9.5.4 Mounting provisions. The RF coupling and mechanical mounting provisions for mounting the antenna on the pedestal shall be such that the antenna can, without modification, be mounted on the pedestal for the AN/UPA-22 antenna system, as shown in FIGURE 6. Mounting provisions shall be as shown in FIGURE 6. Two sets of mounting holes, spaced as shown in FIGURE 6 shall be provided. The design of the antenna shall be such that it will be possible to mount the antenna on a radar antenna structure when proper mounting fixtures are provided.

3.9.6 Broadband IFF antenna system components. The broadband IFF antenna components shall consist of the following component parts as a design minimum. Where component integration is feasible, the design philosophy shall be incorporated.

3.9.6.1 Radiating elements. The radiating element used shall be a broadband design covering 600 MHz to 1100 MHz. The radiating element patterns, gain, VSWR shall be consistent with achieving the system primary requirement specified herein. A yagi array is acceptable for meeting this requirement.

3.9.6.2 Feeder circuits. Two feeder circuits, capable of coupling RF energy from the signal processor unit to the 12 log periodic (LP) antenna elements, shall be provided.

3.9.6.2.1 Electrical. The two feeder circuits to be supplied shall be capable of providing a CHEBYSHEV distribution of the signal amplitudes to the radiating elements. The feeder circuits shall have provisions for adjusting the phases of each individual antenna element and the input to each feeder circuit.

3.9.6.2.2 Mechanical. The two feeder circuits shall be modular in construction, utilizing strip line techniques such that both assemblies can be readily assembled into or disassembled from the system for repair or replacement.

3.9.6.3 Signal processor assembly. The signal processor assembly shall be located on the antenna assembly and shall contain the following components:

a. AN/UPA-61 (Government Furnished Equipment)
b. Diplexer
c. Hybrid
d. Directional Coupler

3.9.6.3.1 Mechanical. The signal processor shall be designed so that any component can be removed without disturbing or removing any other component in the assembly. The signal processor assembly shall be equipped with handle(s) for removing the complete assembly from the broadband antenna.

3.9.6.3.2 Electrical. The signal processor assembly shall have one RF input connector (Type N), three RF output connectors (Type N), and one video input connector (BMC) for the switch driver pulse. Interconnecting cable design shall be such that the signal processor component package can be operational with or without the AN/UPA-61 RF switch. All necessary mating connectors shall be provided as part of the equipment.

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3.9.6.4 Diplexers. The diplexers shall be designed for hard mounting on the IFF antenna for above-decks shipboard environment. The following are the requirements for the diplexers:

a.	Frequency range:	Low band: 600-800 MHz High band: 1000-1100 MHz
b.	Insertion loss:	Low and high pass bands: 0.5 dB maximum Low and high reject bands: >40 dB
c.	Power handling:	High and low band: 5.0 KW peak 25 watts average
d.	VSWR:	≤1.5:1
e.	High pass filters shall have DC continuity to permit passage of a DC pulse (-50 V amplitude) to an RF switch.	
f.	Environmental conditions:	Shall conform to the requirements of 3.1
g.	Weight:	3.1752 kilograms (seven pounds) maximum
h.	Size:	See FIGURE 2 for Type I See FIGURE 3 for Type II
i.	The Type I and Type II diplexers shall be in accordance with FIGURE 2 and FIGURE 3 respectively.	
j.	Connectors:	Type N

3.9.6.5 Hybrid. The following are the requirements for the 180 degree hybrid assembly:

a.	Frequency range:	600 to 1150 MHz
b.	Power handling:	5 KW peak 25 watts average
c.	Isolation:	20 dB minimum 30 dB design objective
d.	Amplitude balance:	±0.5 dB maximum
e.	VSWR:	≤1.5 maximum
f.	Phase:	180 ± 9.0 degrees
g.	Connectors:	Type N
h.	Size:	See FIGURE 4
i.	Weight:	0.4536 kilograms (one pound) maximum

3.9.6.6 Directional coupler. The following are the requirements for the directional coupler:

a.	Frequency range:	600 to 1100 MHz
b.	VSWR:	1.25:1 maximum
c.	Power handling:	5 KW peak 25 watts average
d.	Directivity:	20 dB minimum
e.	Coupling:	Value necessary to achieve ± SLS system performance
f.	Weight:	0.68040 kilograms (1.5 pounds) maximum
g.	Size:	See FIGURE 5

3.9.6.7 Backfill radiator assembly. A backfill radiator assembly shall be provided on the LP antenna to provide coverage of the backlobe. The backfill radiator shall be housed in its own radome and sealed against shipboard environment. The backfill radiator pattern in conjunction with the directional antenna pattern shall provide coverage of the sum pattern sidelobes. It shall be designed to provide at least four dB coverage of all sum pattern sidelobes and backlobes. The backfill radiator shall be designed to operate at 1030 MHz ± 10 MHz only.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; b) Government product inspection to measure quality of the product to be offered for acceptance; c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The methods of inspection are classified as follows:

a. First article inspection (4.3)
b. Quality conformance inspections (4.4)

1. Production inspection (Group A)
2. Production control inspection (Group B)
3. Environmental inspection (Group C)

c. Inspection of preparation for delivery (4.6)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE I.

TABLE I. First article and quality conformance inspections.

Examination or test	Requirement paragraph	Test paragraph	First article	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination						
Weight	3.9.5.1, 3.9.6.4g, 3.9.6.5i, 3.9.6.6f	4.5.1	X	X		
Size	3.9.5.1, 3.9.6.4h, 3.9.6.5h, 3.9.6.6g	4.5.1	X	X		
Parts and Materials	3.1.1, 3.2	4.5.1	X	X		
Finish	3.1.1	4.5.1	X	X		
Marking	3.1.1	4.5.1	X	X		
Safety	3.1.1	4.5.1	X	X		
Workmanship	3.1.1	4.5.1	X	X		
Performance	3.9	4.5.2	X	X		
Temperature	3.4	4.5.3	X			
Low temperature	3.4	4.5.3.1	X		X	X
High temperature	3.4	4.5.3.2	X		X	X
Humidity	3.5	4.4.3	X			X
Wind velocity	3.1.1b	4.5.5	X			
Icing	3.1.1c	4.5.6	X			
Interchangeability	3.1.1h	4.5.9	X			X
Nuclear Air Blast	3.1.1e	4.5.8	X			
Shock	3.1.1f	4.4.3	X			X
Vibration	3.1.1g	4.4.3	X			X
Magnetic field environment		4.5	X			X
Magnetic		4.5	X		X	
Thermal performance test	3.8	4.5.10	X			

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4.4 Quality conformance inspections.

4.4.1 Production inspection (Group A). Production inspection shall be made on every equipment offered for delivery. Production inspection shall comprise such examinations and tests as will prove the workmanship and reveal omissions and errors of the production process, such as a) functional and performance test at a limited number of points in the required range, b) tests which detect deviations from design, tests of controls and adjustments, and c) tests which detect hidden defects of materials. Production inspection shall include the examination and tests listed in the Group A column of TABLE I.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. This inspection shall consist of the examinations and tests which encompass a) functional and performance tests throughout the entire range of operation, b) tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, or by substitution of different parts, c) tests which will detect deviations in the processing of the materials, d) tests to determine temperature rise produced in operation and ability of equipment to withstand this heat, e) tests of efficiency, and f) tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examinations and test shown in the Group B column of TABLE I.

4.4.2.1 Sampling plan for Group B inspection. Equipments submitted for Group B sampling tests shall be selected by the contractor under the supervision of the Government Quality Assurance Representative and shall be representative of current production. Group B inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on inspection lots that have passed Group A inspection. The acceptable quality level (AQL) shall be 6.5 per cent defective and the inspection level shall be S-3 for normal, tightened, and reduced inspection.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lo , using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspections shall be conducted on a sampling basis as specified in 4.4.3.1. This inspection shall encompass environmental tests to prove a) the durability of the materials, parts, units, and the equipment as a whole, b) life tests and simulated service tests, c) test of the effects of changes of environment (such as extremes of temperature and humidity, effect of salt air), and d) tests of the effects of shock, vibration, and inclination. Environmental inspection shall include the examinations and tests shown in the Group C column of TABLE I.

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments, or fraction thereof, produced. The first sample equipment shall be selected from the first months production.

4.4.3.1.1 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3.1, the contractor shall immediately investigate the cause of failure and shall report to the Quality Assurance Representative the results thereof, and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. The equipment shall then be retested.

4.4.4 Reinspection of conforming production control and environmental test units. Unless otherwise specified, sample units which have been subjected to and passed production control or environmental inspections may be accepted on the contract, provided they are resubjected to and pass production inspection after repair of all visible damage.

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4.4.5 Test equipment and test conditions. Test equipment and test conditions for first article inspection and quality conformance inspection shall comply with the following quality assurance provisions of MIL-E-16400:

a. Test facilities, chambers, and apparatus
b. Standard test conditions
c. Performance of test

4.5 Test methods. Unless otherwise specified herein, the inspections listed in TABLE I shall be performed in accordance with the test methods specified in MIL-E-16400.

4.5.1 Surface examination. The equipment shall be examined to ensure compliance with the specified requirements for:

a. Weight (3.9.5.1, 3.9.6.4g, 3.9.6.5i, 3.9.6.6f)
b. Size (3.9.5.1, 3.9.6.4h, 3.9.6.5h, 3.9.6.6g)
c. Parts and materials (3.1.1, 3.2)
d. Finish (3.1.1)
e. Marking (3.1.1)
f. Safety (3.1.1)
g. Workmanship (3.1.1)

In the process of examination, the equipment shall not be disassembled in a manner that will affect the performance, durability, or appearance of the unit. The examination shall include a check of all operating controls, circuit functions, test provisions, and adjustments.

4.5.2 Performance test. The equipment shall be subjected to a performance test, including all operating controls, to determine compliance with the performance parameters and safety requirements of 3.9. The performance test shall include the individual tests of TABLE II.

TABLE II. Performance tests.

Test	Requirement paragraph	Test paragraph
Antenna system component tests:		4.5.2
Diplexers (insertion loss, power handling, VSWR, and DC continuity)	3.9.6.4	
Hybrid (amplitude balance, VSWR, power handling, and isolation)	3.9.6.5	
Directional coupler (directivity, VSWR, and power handling)	3.9.6.6	
Complete antenna system tests:		4.5.2
Power handling capability	*3.9.1	
Gain and VSWR	3.9.2, 3.9.3.6	
Horizontal antenna patterns	3.9.3.1, 3.9.3.2	
Vertical antenna patterns	3.9.3.3, 3.9.3.4, 3.9.3.5	
*First article only.		

4.5.3 Temperature.

4.5.3.1 Low temperature. The low temperature test shall be in accordance with Method 502 of MIL-STD-810, except as follows:

a. Step 2: Storage or non-operating temperature shall be -62°C $^{+0°C}_{-5°C}$ for a duration of 24 hours.
b. Step 4: Temperature range for above-deck equipment shall be -28°C
Temperature range for below-deck equipment shall be 0°C
c. The combined duration of Steps 4 and 5 shall be not less than 12 hours, at least two of which shall be in the stabilized condition prior to conducting tests.

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4.5.3.2 High temperature. The high temperature test shall be in accordance with Method 501, Procedure I, of MIL-STD-810, except as follows:

a. Step 4: Highest operating temperature shall be +150°C for above deck equipment and +50°C for below deck equipment
b. The combined duration of Steps 4 and 5 shall be not less than 12 hours, at least six of which shall be in the stabilized condition prior to conducting tests.
c. Step 7: The equipment shall operate for a period of at least two hours in the stabilized condition before the Step 7 tests are performed.

4.5.4 Humidity. Humidity tests shall be in accordance with Method 507, Procedure IV of MIL-STD-810, except as follows:

a. Step 5: Equipment performance measurements for temperature Range 4 shall be made at 50°C (122°F).
b. Step 9: The equipment shall be carefully examined in detail to detect evidence of physical degradation. Blistering of organic coatings shall be a cause of rejection.

4.5.5 Wind velocity. The above deck equipment, installed in the test facility in a manner that will simulate service usage, shall be subjected to a suitable wind test to demonstrate compliance with 3.1.1b.

4.5.6 Icing. The exposed equipment shall be subjected to a test simulating the icing environment to demonstrate conformance to 3.1.1c.

4.5.7 Gunblast. Gunblast test shall be conducted as a simulated test wherein the item shall be mounted on the carriage of the US Navy simulated gunblast equipment. The front edge of the item shall be positioned in the test plane, and the axis shall be coincident with that of the explosion chamber. The item being tested shall be subjected to 30 rounds of blast at a peak pressure of 9.5 lb/in^2 g.

4.5.8 Nuclear air blast. The ability of the above-deck equipment to withstand the nuclear air blast environment of 3.1.1e shall be verified by nuclear air blast calculations.

4.5.9 Interchangeability. The antenna system shall be tested to demonstrate compliance to 3.1.1h by utilizing at least two of each interchangeable subsystems and antenna components for each demonstration. Interchangeability of the antenna base mounting and alignment hole pattern and hole diameters shall be verified by gaging or measurement.

4.5.10 Thermal performance test. In conjunction with first article temperature testing, a thermal performance test shall be performed in accordance with MIL-E-16400. The system shall be operating in an ambient temperature of 150°C with a duty cycle as specified in 3.9.1 during the test.

4.6 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

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6. NOTES

6.1 Intended use. The broadband IFF antenna system is intended to operate in two bands frequency, 1030 MHz ± 10 MHz and 1090 ± 10 MHz, and low frequency, from 600 MHz to 800 MHz). igh frequency band operates with sum and difference radiated patterns in an ISCS mode at MHz ± 10 MHz. The low frequency band and the 1090 MHz ± 10 MHz operate in the sum pattern

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. Levels of preservation, packaging, packing, and marking (see 5.1)

Preparing activity:
NAVY-EC
(Project 5985-N455)

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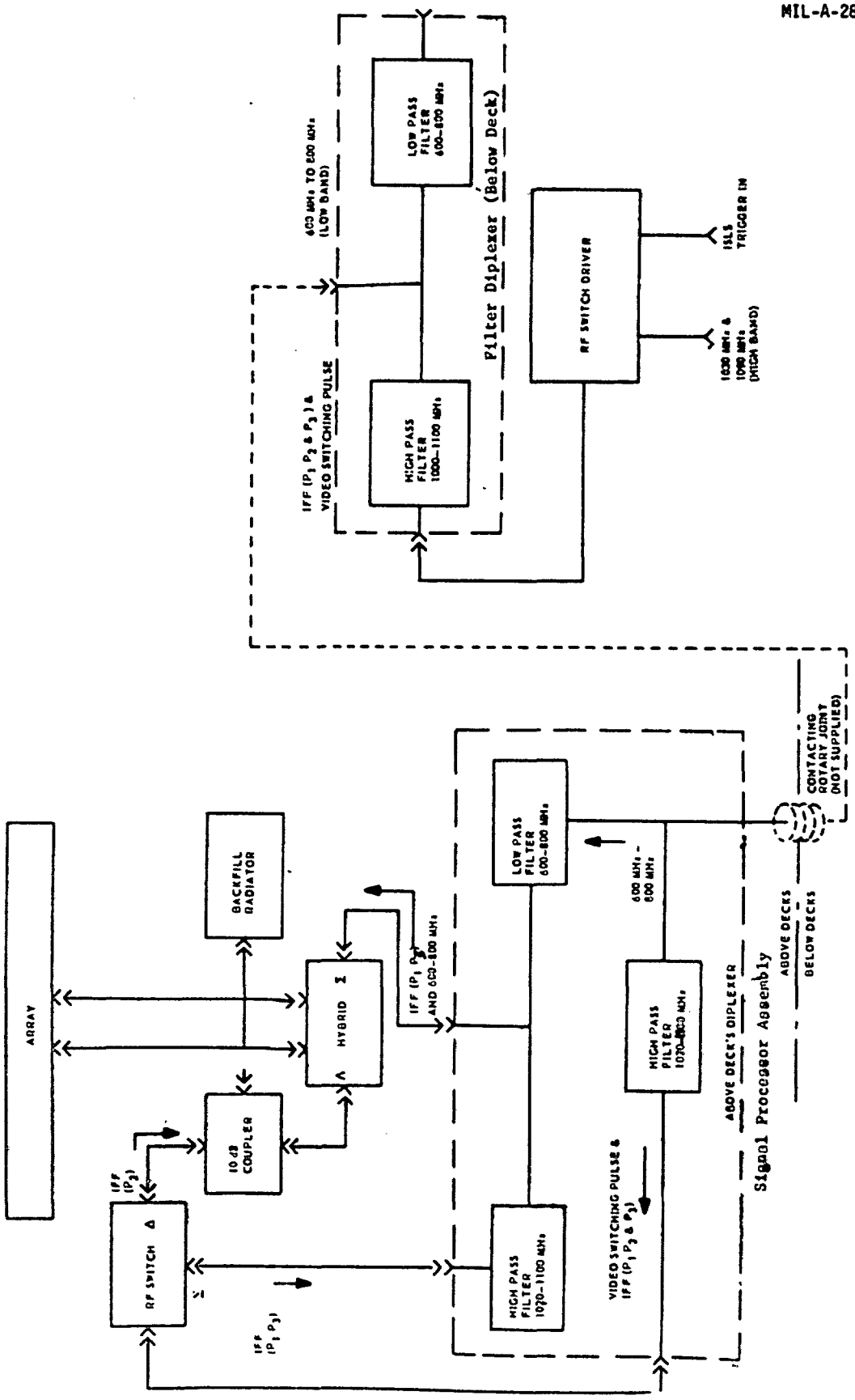
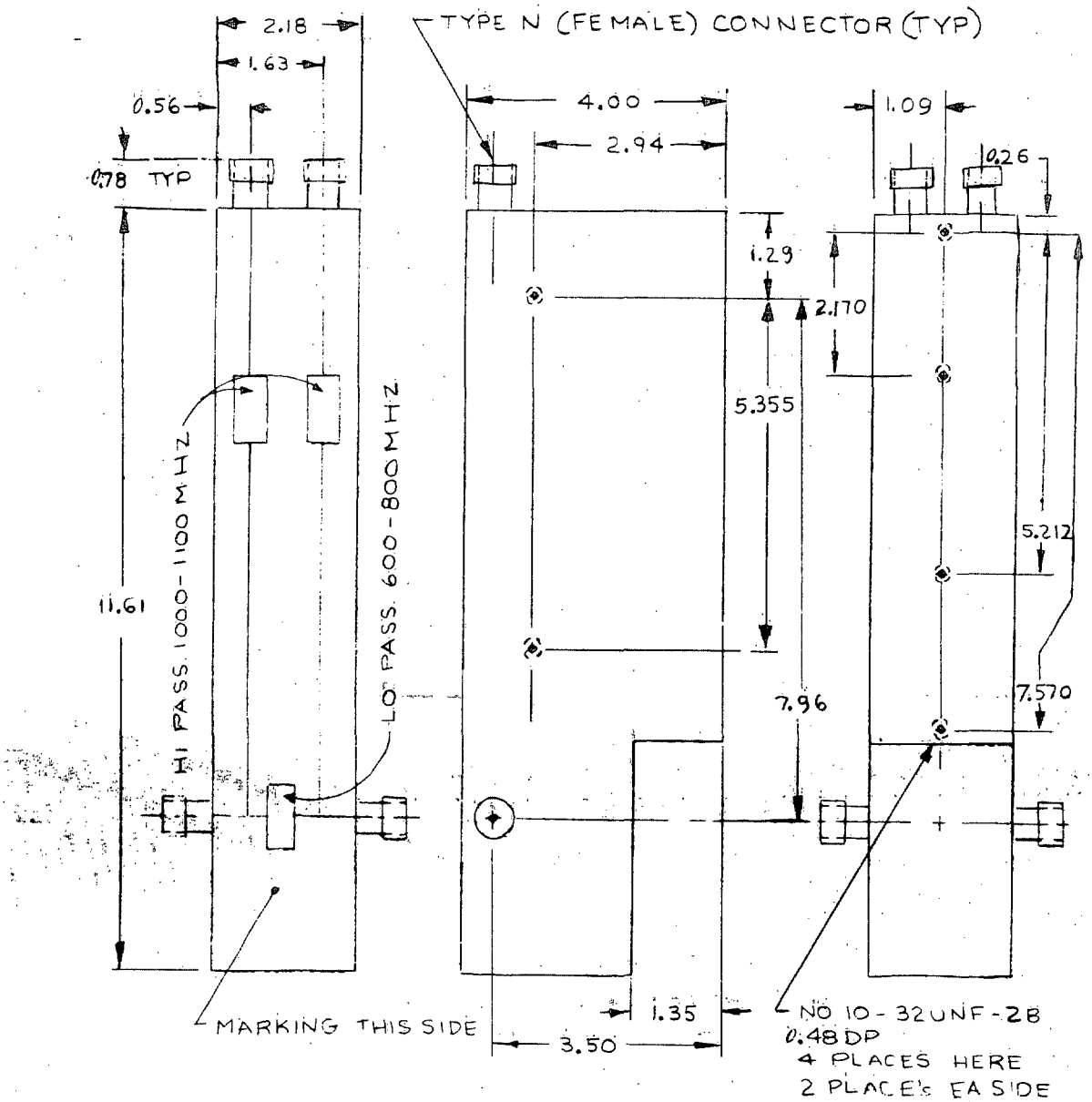


FIGURE 1. Broadband IFF antenna system block diagram.

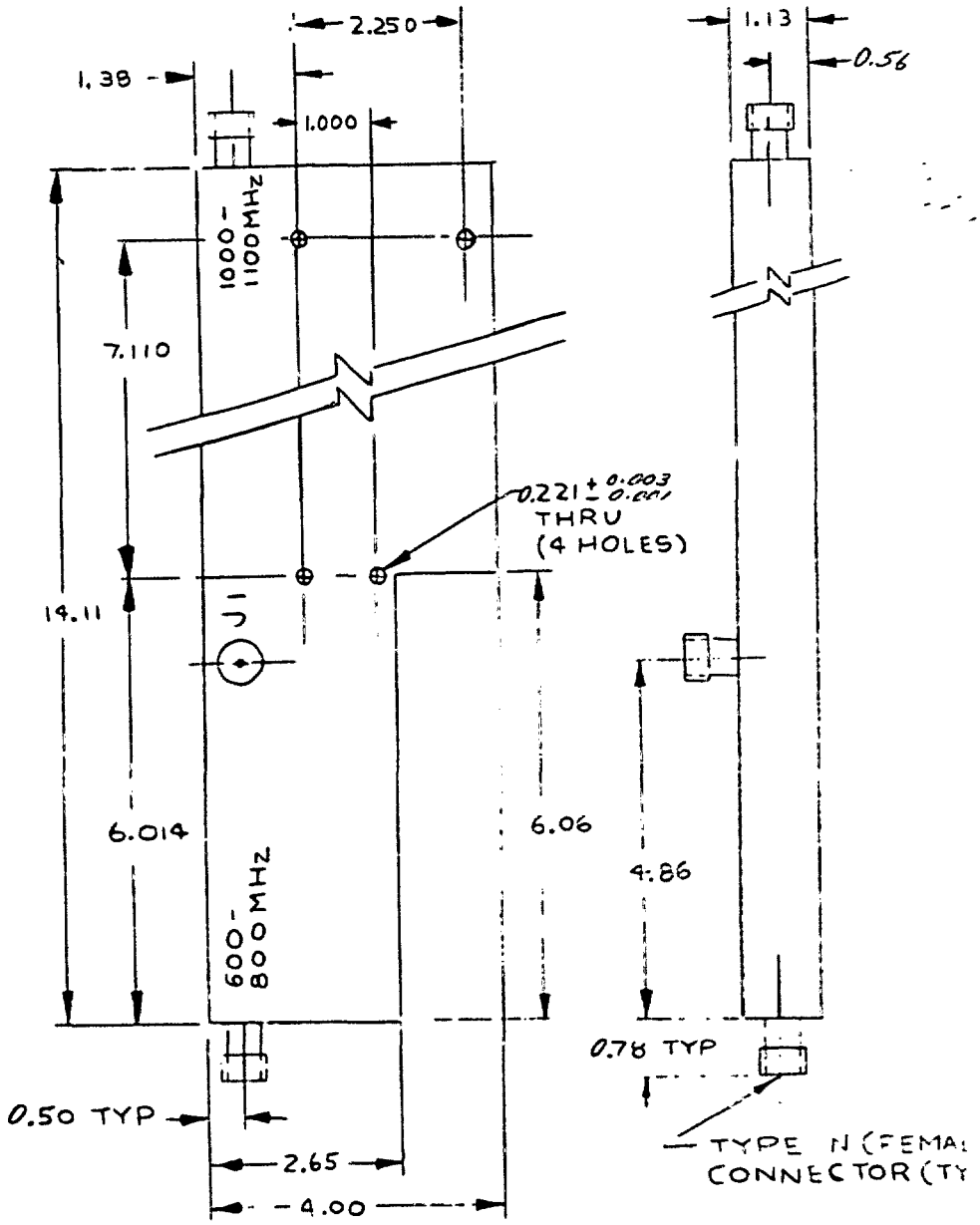
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Metric equivalents			
Inches	millimeters	Inches	Millimeters
0.26	6.60	2.170	55.12
0.48	12.192	2.18	55.37
0.56	14.22	2.94	74.68
0.78	19.81	3.50	88.90
1.09	27.69	4.00	101.60
1.29	32.77	5.212	132.39
1.35	34.29	5.355	136.02
1.63	41.40	7.570	192.28
		7.96	202.18

FIGURE 2. Type I diplexer.

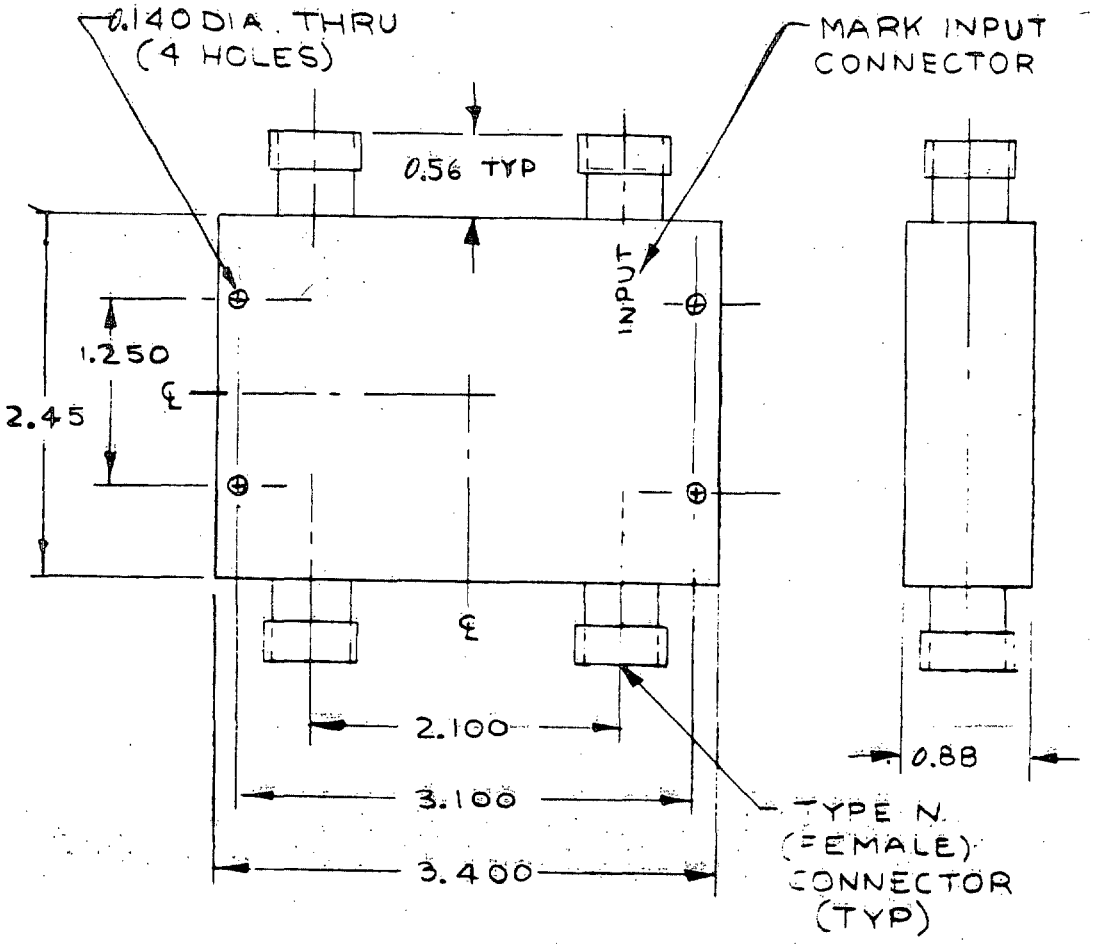
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:16Z
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Metric equivalents.					
Inches	Millimeters	Inches	Millimeters	Inches	Millimeters
0.001	0.025	1.000	25.40	4.86	123.44
0.003	0.076	1.13	28.702	6.014	155.960
0.221	5.6134	1.38	35.05	6.06	167.640
0.50	12.70	2.250	57.150	7.110	180.594
0.56	14.22	2.650	67.31	14.11	358.394
0.78	19.81	4.00	101.60		

FIGURE 3. Type II diplexer.

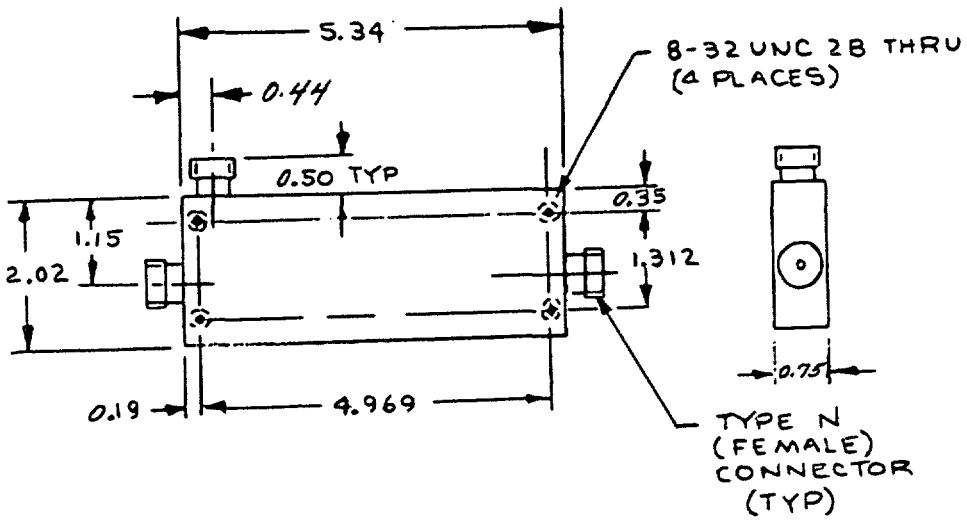
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:16Z
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Metric equivalents.			
Inches	Millimeters	Inches	Millimeters
0.140	3.56	2.100	53.34
0.56	14.22	2.45	62.23
0.88	22.35	3.100	78.740
1.250	31.750	3.400	86.36

FIGURE 4. Hybrid assembly.

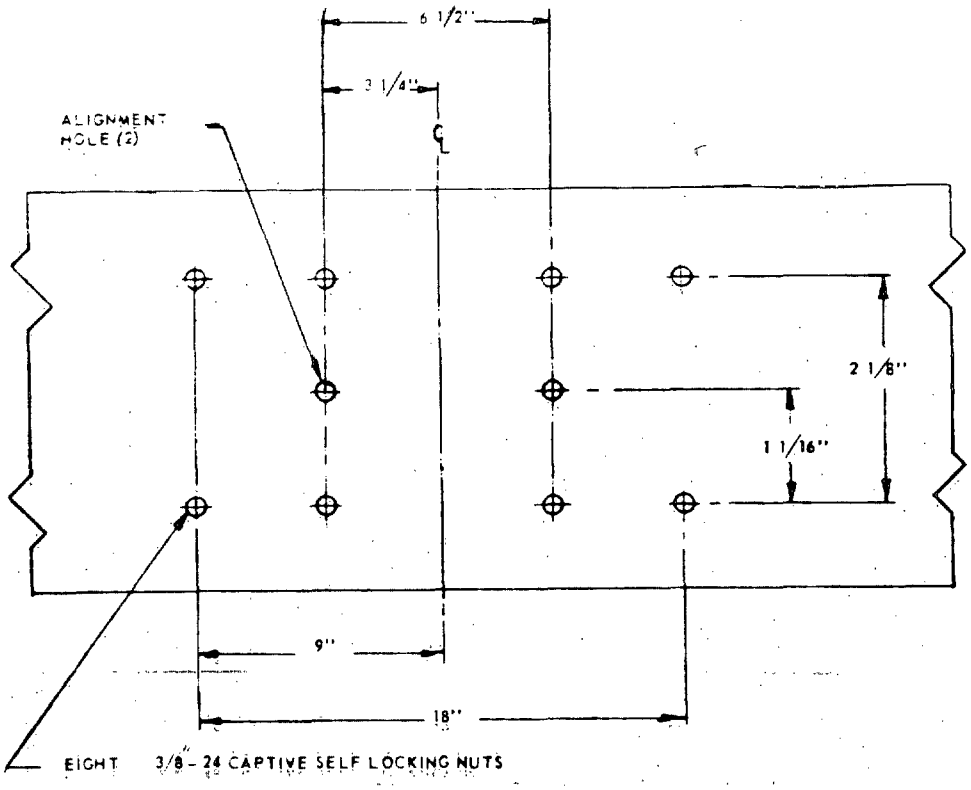
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Metric equivalents.	
Inches	Millimeters
0.19	0.4826
0.35	8.890
0.44	11.18
0.50	12.70
0.75	19.05
1.15	29.210
1.312	33.3248
2.02	55.880
4.969	126.2126
5.34	135.640

FIGURE 5. Directional coupler.

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Metric/Decimal equivalents.		
Inches		Millimeters
Fraction	Decimal	
1-1/16	1.0625	26.9875
2-1/8	2.125	53.975
3-1/4	3.250	82.550
6-1/2	6.500	165.100
	9.000	228.600
	18.000	457.200

FIGURE 6. Base mounting holes.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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